

University Interscholastic League
2023 – 2024 Junior High Number Sense Test A

Contestant's Number _____

Final		
2 nd		
1 st		
	Score	Initials

**Read Directions Carefully
Before Beginning Test**

**Do Not Unfold This Sheet
Until Told to Begin**

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. **ALL PROBLEMS ARE TO BE SOLVED MENTALLY.** Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.

Stop – Wait for Signal!

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| <p>(1) $413 + 51 =$ _____</p> <p>(2) $5 \times 16 =$ _____</p> <p>(3) $88 - 49 =$ _____</p> <p>(4) $111 \div 3 =$ _____</p> <p>(5) $\frac{11}{16} - \frac{1}{4} =$ _____</p> <p>(6) $12 \div 0.4 =$ _____</p> <p>(7) $45 \times 11 =$ _____</p> <p>(8) $14^2 =$ _____</p> <p>(9) $24 - 16 \div 4 =$ _____</p> <p>*(10) $2023 \times 99 + 2024 =$ _____</p> <p>(11) $18 \times 22 =$ _____</p> <p>(12) If the arithmetic mean of 8, x, and 20 is 15, $x =$ _____</p> <p>(13) Which is larger: $\frac{5}{11}$ or $\frac{6}{13}$? _____</p> <p>(14) $15 \times 35 + 5 \times 35 =$ _____</p> <p>(15) $48 \times 68 =$ _____</p> <p>(16) $100 - 32 - 18 =$ _____</p> <p>(17) $33^2 =$ _____</p> <p>(18) $28 + 32 + 36 + 40 =$ _____</p> <p>(19) MMDXL = _____ (Arabic Numeral)</p> | <p>*(20) $18.75 \times 320 =$ _____</p> <p>(21) $18 \times 22 + 4 =$ _____</p> <p>(22) $101 \times 567 =$ _____</p> <p>(23) If $f(x) = -2x^2 - 15$ then $f(-4) =$ _____</p> <p>(24) $5\frac{1}{11} \times 6\frac{1}{11} =$ _____ (mixed number)</p> <p>(25) $4\frac{11}{12} + 6\frac{5}{6} =$ _____</p> <p>(26) $31 - 31^2 =$ _____</p> <p>(27) The positive square root of 625 is _____</p> <p>(28) If n is to 6 as 8 is to 3, then $n =$ _____</p> <p>(29) If $20 + 4x$ is 44, then $x =$ _____</p> <p>*(30) $229^2 =$ _____</p> <p>(31) $82 \times 88 =$ _____</p> <p>(32) The sum of the prime numbers less than 20 but greater than 11 is _____</p> <p>(33) If $1.75 + 0.5 = n$, the $n^{-1} =$ _____</p> <p>(34) 24 is divisible by how many whole numbers? _____</p> <p>(35) 3 square yards = _____ square feet</p> <p>(36) 6 percent = _____ (common fraction)</p> <p>(37) If the sales tax is $8\frac{1}{2}\%$, what is the amount of sales tax on an item that costs \$50? \$ _____</p> |
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- (38) The LCM (24, 15) $\times$ GCF (24, 15) = _____
- (39) The area of an equilateral triangle with a side of 8-cm is $a\sqrt{3}$ and a = _____ cm²
- *(40) $1111\sqrt{9801}$ = _____
- (41) 63 (base 10) = _____ (base 5)
- (42) $5^4 + 5^3$ = _____
- (43) 649×111 = _____
- (44) What is the area of a rhombus with diagonals 120 and 100? _____
- (45) 32 (base 5) $- 4$ (base 5) = _____ (base 5)
- (46) If $30 \div 3 = n^2 + 1$, and $n > 0$, then n = _____
- (47) If $A = \{2, 3, 4, 5\}$, $B = \{1, 2, 4, 6\}$ and $C = \{2, 4, 6\}$, how many elements are in $A \cap C \cap B$? _____
- (48) 18 pints = _____ quarts
- (49) The surface area of a rectangular solid that measures 8 by 5 by 2 is _____
- *(50) 625×160 = _____
- (51) In the sequence: 1, 4, a , 16, 25, b , 49, . . .
 $(a - b)^2$ = _____
- (52) What is the 6th term in the sequence: 1, 4, 9, . . .? _____
- (53) The sum of the interior angles for a square is _____ $^\circ$
- (54) What is the area of a parallelogram with opposite sides each 16 and altitude between them 25? _____
- (55) What is the area of a rectangle with a perimeter of 24 and length 8? _____
- (56) What is the radius of a circle with a circumference of 100π ? _____
- (57) The hypotenuse for a triangle with sides 8, 15 is _____
- (58) $(15^2 + 8 \times 6) \div 4$ has a remainder of _____
- (59) The surface area for a cube with edge 5 is _____
- *(60) $15^4 \times 111 \div 5^3$ = _____
- (61) $(25^{-1} \times 10^2)^{-1}$ = _____
- (62) 440 yards = _____ feet
- (63) 36 km/hr = _____ meters/sec
- (64) $8\frac{1}{3}\% + 33\frac{1}{3}\%$ = _____ (common fraction)
- (65) $25^3 \div 7$ has a remainder of _____
- (66) $\frac{1+2+3+\dots+7}{1+3+5+\dots+9}$ = _____
- (67) The sum of the positive divisors of 20 is _____
- (68) $4!$ = _____
- (69) $5\frac{1}{4} \div \frac{3}{8}$ = _____
- *(70) 15 miles = _____ yards
- (71) $11^2 + 33^2$ = _____
- (72) $55\frac{5}{9}\%$ = _____ (common fraction)
- (73) $0.1333\dots$ = _____ (common fraction)
- (74) The surface area of a sphere with a radius of 11 is $k\pi$ and k = _____
- (75) What are the odds of randomly picking a red ball from a bag of 12 red and 18 blue balls? _____
- (76) $24^2 \times 25$ = _____
- (77) If the shortest distance between the points (0, 7) and (x , 0) is 25, what is x ? _____
- (78) $12 + \frac{5!}{3!}$ = _____
- (79) $(9.5)^2 - (5.5)^2$ = _____
- *(80) 360 days = _____ hours

2023 – 2024 University Interscholastic League Junior High Number Sense Test A – Key

(1) 464	*(20) 5700 – 6300	(38) 360	(59) 150
(2) 80	(21) 400	(39) 16	*(60) 42708 – 47202
(3) 39	(22) 57267	*(40) 104490 – 115488	(61) $\frac{1}{4}$; .25
(4) 37	(23) -47	(41) 223	(62) 1320
(5) $\frac{7}{16}$; .4375	(24) $31\frac{1}{121}$	(42) 750	(63) 10
(6) 30	(25) $11\frac{3}{4}$; 11.75; $\frac{47}{4}$	(43) 72039	(64) $\frac{5}{12}$
(7) 495	(26) -930	(44) 6000	(65) 1
(8) 196	(27) 25	(45) 23	(66) $\frac{28}{25}$; $1\frac{3}{25}$; 1.12
(9) 20	(28) 16	(46) 3	(67) 42
*(10) 192186 – 212416	(29) 6	(47) 2	(68) 24
(11) 396	*(30) 49819 – 55063	(48) 9	(69) 14
(12) 17	(31) 7216	(49) 132	*(70) 25080 – 27720
(13) $\frac{6}{13}$	(32) 49	*(50) 95000 – 105000	(71) 1210
(14) 700	(33) $\frac{4}{9}$	(51) 729	(72) $\frac{5}{9}$
(15) 3264	(34) 8	(52) 36	(73) $\frac{2}{15}$
(16) 50	(35) 27	(53) 360	(74) 484
(17) 1089	(36) $\frac{3}{50}$	(54) 400	(75) $\frac{2}{3}$
(18) 136	(37) 4.25	(55) 32	(76) 14400
(19) 2540		(56) 50	(77) 24
		(57) 17	(78) 32
		(58) 1	(79) 60
			*(80) 8208 – 9072

Note: *(Number) x – y means an integer between x and y inclusive.

If an answer is of the type like $\frac{2}{3}$ it cannot be written as .666... or $\overline{.6}$.